education sciences

MDPI

Article

Factors Influencing University Teachers' Technological Integration

Judit T. Nagy [1,*] and Ida Dringó-Horváth [2]

[1] Department of Sociology, Faculty of Humanities and Social Sciences, Károli Gáspár University of the Reformed Church in Hungary, 1088 Budapest, Hungary
[2] ICT Research Center, Educational Technology Training Centre, Károli Gáspár University of the Reformed Church in Hungary, 1091 Budapest, Hungary; dringo.horvath.ida@kre.hu
* Correspondence: tnagy.judit@kre.hu

Abstract: The extent and effectiveness of digitalization are influenced by a variety of factors, which are worth studying both separately and together, looking at how they affect each other. Typically, researched aspects in the context of education are institutional support, teachers' beliefs about using digital tools, digital competence in teaching, technostress, self-efficacy, and the extent of digital tool use. The present study aims to center around the final aspect. It examines the separate influencing factors of information and communications technology (hereafter, ICT) use within a complex model, with the help of partial least squares structural equation modeling. The study was based on a survey of university lecturers (n = 116) at a university in Hungary that has six different training locations and five faculties. It was conducted by using an online questionnaire. To summarize the identified correlations, it can be concluded that digital competence in teaching, collegial support, ICT self-efficacy and ICT perception play a significant role in the use of technology. The effects detailed in the results can contribute to the effective digitalization of higher education institutions in several practical ways.

Keywords: information and communication technology; technostress; higher education; digital competence; self-efficacy; collegial support

check for updates

Citation: Nagy, J.T.; Dringó-Horváth, I. Factors Influencing University Teachers' Technological Integration. *Educ. Sci.* **2024**, *14*, 55. https://doi.org/10.3390/educsci14010055

Academic Editor: Ana García-Valcárcel

Received: 17 November 2023
Revised: 1 December 2023
Accepted: 2 December 2023
Published: 2 January 2024

Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

In most European countries, including Hungary, the ubiquitousness of digitalization and its impact on all aspects of higher education are mainly a consequence of the COVID-19 pandemic in 2020. The period of forced digital transition has demonstrated that higher education institutions can ensure continuity of teaching and learning, but it has also shown that there is still much to be done to make effective use of digital technologies in higher education for quality assurance, inclusion, and equity [1].

The importance of the topic in Hungary is demonstrated by the fact that—as in other countries—more and more initiatives have been launched to map and research the digital transformation of higher education, and to support the effective further development of the process [1].

The extent and effectiveness of digitalization is influenced by a variety of factors, which should be considered both individually and collectively, taking into account their impact on each other. Factors typically examined in the context of education include institutional support, educators' beliefs about digital tools, digital competence of educators, technostress, self-efficacy, and the extent of tool use.

This study aims to focus on the integration of ICT information and communications technology (hereafter, ICT) in university education by exploring the main factors that support and hinder the use of technology.

Research to date suggests that inadequate technological competence (i.e., lack of training), lack of infrastructure (i.e., lack of technology and IT support), and limited

resources are the most important factors that prevent teachers from effectively integrating technology into their teaching practice [2,3]. Moreover, teachers' attitudes/beliefs are key determinants that reflect their behavior when using technology in their teaching practice [4].

Some aspects influencing the integration of ICT in education, such as teachers' professional knowledge as an individual factor, institutional support as an environmental factor, teachers' convictions about technology, and the interplay of all these, have been investigated in several studies for public education [2,5]. However, no research has been published on the complex, combined analysis of these factors for higher education.

Higher education-related studies have only investigated other factors and their interrelationships: Taimalu and Luik [6] examined the impact of teacher educators' beliefs and professional knowledge on their use of technology, while Luik et al. [7] conducted a teacher education-related study on teacher trainees using the TPACK model to measure their perceived level of technological, pedagogical and professional knowledge, and the impact of all these on technology use.

Considering all the above-mentioned points, establishing a complex, structural model that reveals the factors influencing the use of ICT by university lecturers and their correlations is relevant and of paramount importance. The results could contribute to a significant increase in ICT use, with tangible practical benefits for the effective digitalization of higher education.

2. Theoretical Background

In this section, some elements of the research model will briefly be presented, outlining the research findings related to them. It should be noted that in the literature review, we have not only used literature and research related to university teachers, but to teachers in general.

2.1. ICT Use

Although several studies have examined the use of ICT by teachers in higher education [8–11], no study has come to our attention that explores the factors influencing use or integration together. Studies of influencing factors have mostly focused on behavior intention, based on the view that intention to use covers use.

In this research, we focus directly on ICT use, by which we mean the frequency and extent of (self-reported) use of ICT for teaching and research purposes by higher education teachers, i.e., the range of ways in which higher education teachers use these tools, based on the six domains of the Digital Competence of Educators (DigCompEdu) framework (see Section 2.3 for more details on the framework).

In the studies that emerged after the 2020 COVID-19 pandemic, and that focused mainly on the experience of forced remote education, the use of tools in education was much more strongly represented than before, either looking at the whole process [12] or a sub-area, such as the use of tools in relation to assessment [13]. In addition, there are studies related to the need for and possible forms of continuous professional development related to tool use [14,15].

Improving the use of ICT among teachers in higher education is essential to enhance teaching and learning experiences, adapt to technological advancements, promote active and autonomous learning, increase accessibility and inclusivity, enhance efficiency and productivity, and foster a culture of lifelong learning. By embracing ICT, teachers can effectively prepare students for the challenges and opportunities of the digital age. Additionally, an in depth systematic review carried out using 100 research articles on academic performance and ICT use results, there was evidence of improved performance in educational practices enriched with ICT [16]. All this justifies the choice of this area as the main focus of this study.

2.2. Technostress

In encouraging teachers to use ICT more actively, we should not overlook the psychological and physical stress associated with technology use, which Brod defined in 1984 as technostress: "a modern disease of adaptation caused by an inability to cope with new computer technologies in a healthy manner" [17]. Weil and Rosen put it this way [18]: "any negative impact on attitudes, thoughts, behaviors, or body physiology that is caused either directly or indirectly by technology".

Previous research has shown that there are a number of factors that can cause technostress, such as system failure, insufficient technical and social support in the use of technology, increased time spent on class preparation, and a negative presupposition in educational institutions related to the use of technology [2,19].

Studies in educational contexts have shown that teachers' anxiety and stress about technology have a negative impact on their motivation and intention to use ICT [20–22]. They also have a negative impact on teachers' actual use of digital tools [2,23,24].

Based on the above literature review, it can also be stated that in order to effectively reduce teacher technostress, teachers must have internal resources to assess the event or situation and external resources to assess their ability to handle the event or situation. The former is called individual factor, and the latter, environmental factor.

Teacher technostress is, therefore, influenced by both individual and environmental factors, all of which need to be examined individually to provide appropriate support in reducing it. In this study, digital pedagogical competence is examined as an individual factor, and institutional support as an environmental factor. These two will be discussed in detail below, followed by the analysis of an additional aspect of the research model, namely, beliefs.

2.3. Digital Pedagogical Competence

The digital competences of teachers are considered to be a combination of professional, pedagogical and technological knowledge and skills [25], and are defined in more detail as including all "skills related to the use of ICT in teaching and learning as well as other educational activities (instructional management, related individual and organizational communication, research activities)" [26]. A number of models have been developed for assessing and classifying teachers' digital competences, such as TPACK model, SAMR model, Technological Integration Matrix (TIM), RAT model, PICRAT model and DigCompEdu framework [27], which is the educational version of the DigComp framework, previously developed by the European Commission. Since it is somewhat more widespread in Hungary and there is also a version specifically optimized for higher education, in this study we used the DigCompEdu [28] framework to measure digital competence in education. The framework is general enough to be used in different educational contexts [28], with the advantage that there is a higher education-specific version.

The negative impact of digital pedagogical competence on teachers' technostress, which has been confirmed by several studies [2,5,19,29], is significant because it ensures that teachers' stress can be reduced by teaching them to use technology appropriately. Thus, by developing digital pedagogical competence, we can support teachers' coping mechanisms with technology-induced psychological stress, and, in turn, positively influence technology use.

The nature, complexity and personalization of institutional support can play a very important role in this process.

2.4. Institutional Support

Institutional support, as the main environmental factor, plays an important role in promoting teachers' use of ICT and includes support from the institution for the use of technology for teaching (such as ensuring access to technology, providing digital resources, policy incentives, technical assistance, and encouragement) [30–32].

In this research, we define institutional support as the institutional support perceived by teachers, and distinguish three subcomponents of it, as proposed by Zhao et al. [33]: (1) human infrastructure, (2) technological infrastructure, and (3) social support. The first two components are referred to in the literature as university support, and the third as collegial support.

Technostress can be significantly reduced through institutional support [2,19,34–36], thus, positively influencing technology use.

Several studies have investigated the direct effect of institutional support on teachers' technology use and found that it significantly promoted teachers' technology use [2,37–39]. Some studies have also shown the direct effect of institutional support on digital pedagogical competence [21,40,41].

While the pandemic has acted as a catalyst for digitalization processes and has indeed triggered complex, systemic changes in higher education, which certainly have an impact on the level of institutional support, an OECD 2021 survey reported that more support is needed in this area: only around 40% of respondents agreed that their institution provides them with the opportunity to develop digital skills specific to their field of education and research [1].

With that being said, it seems that university teachers' own beliefs play an even bigger role than institutional support in the success of digitalization.

2.5. Teacher Beliefs

The use of technology in education has been established to be related to teachers' beliefs [42–45]. Liu et al. [46] pointed out that not all technology adoption models incorporate pedagogical beliefs, despite the fact that they are known to be crucial factors for the successful use of technology in education.

When examining the integration of technology in teaching and learning, the most important beliefs are ICT self-efficacy and ICT perceptions, i.e., beliefs about the value of using technology [6], which need to be taken into account to achieve effective use [47].

Self-efficacy is defined by Bandura [48] as an individual's belief in their own abilities to perform certain behaviors or to successfully complete certain tasks. Based on this, the concept of ICT self-efficacy can be defined as teachers' belief and confidence in their ability to use technology effectively to achieve educational goals [47].

Self-efficacy, according to social cognitive theory, strongly influences an individual's behavior towards performing a task, i.e., their ability and willingness to act (whether to make an attempt, how much effort to exert, whether to persist in performing the task [49]. The self-efficacy theory also suggests that emotional reactions are influenced as well (including stress and anxiety [48]).

Similar results have been found in educational contexts: teachers' ICT self-efficacy has been shown to be a critical factor influencing their behavior towards technology integration [5,50–53], their perceived level of technostress [5,24], in addition to it being a strong predictor of their digital pedagogical competence [5,47,54]. In other words, if teachers have higher ICT self-efficacy, it is likely to indicate higher digital pedagogical competence [55].

Several studies have also investigated the mediating role of ICT self-efficacy and found that ICT self-efficacy mediates the effect of institutional ICT support on digital pedagogical competence [5,47], i.e., if there is more institutional ICT support, ICT self-efficacy will have a greater effect on digital pedagogical competence, while if institutional ICT support is lower, it will have a smaller effect. However, the direct effect of institutional ICT support on digital pedagogical competence was not found to be significant [5].

The results of Dong et al. [5] also showed that the indirect effect of ICT self-efficacy on technostress (through digital pedagogical competence) is higher than the directly detectable effect. This interesting finding further reinforces the key role of digital pedagogical competence in reducing teachers' stress during ICT use.

The other component in this area is ICT perceptions, i.e., beliefs about the positive role and value of digital tools in education. The concept summarizes the attitudes, habits and beliefs of educators about the integration of technology in educational activities [56]; it shows the extent to which they value ICT in education [47].

ICT perceptions play a key role in the integration of technology into teaching: positive ICT perceptions increase the likelihood of teachers' use of ICT for teaching purposes, while negative perceptions hinder use [4,44,46,54,57,58].

In addition, previous research has shown the positive impact of teachers' perceptions of ICT on digital pedagogical competence [2,6,47]. Evidence seems to show that positive ICT perceptions of ICT are a prerequisite for teachers to enhance their ICT competence [59]. Moreover, results by Dong et al. [5] show that ICT perceptions have an indirect effect on technostress through influencing digital pedagogical competence.

The mediating role of ICT perception, similar to that of ICT self-efficacy, was investigated by Wang and Zhao [47] in their research, and they concluded that ICT perception mediates the impact of university ICT support on digital pedagogical competence. Another interesting finding was that the direct effect of university ICT support on digital pedagogical competence is not significant. Thus, university ICT support has a positive effect on digital pedagogical competence only through influencing teachers' ICT perceptions.

3. Research Model

The main objective of the study is to examine the factors influencing university teachers' use of ICT and to explore the structural relationships between them by building on the research findings presented in the literature review. This will enable institution- and individual-specific teacher development and, thus, more effective use of ICT in higher education.

To build the model, we considered individual factors (technostress—TS, digital pedagogical competence—DC), belief factors (ICT self-efficacy—ISE, and ICT perception—IP) and environmental factors (university support—US, and collegial support—CS). According to the assumed model (Figure 1), we formulated 3 research questions and 14 hypotheses.

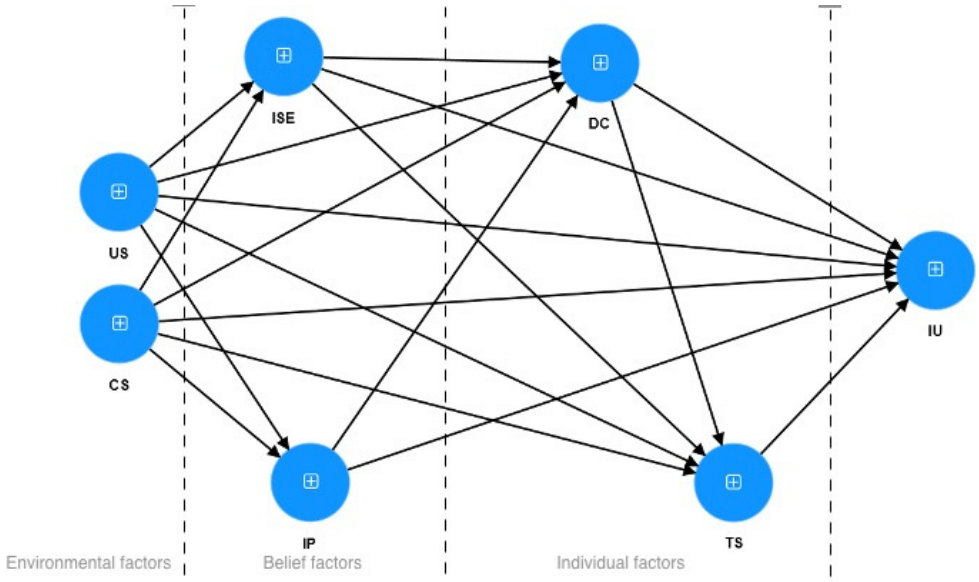


Figure 1. Research model. IU = ICT use; TS = technostress; DC = digital pedagogical competence; US = university support; CS = collegial support; ISE = ICT self-efficacy; IP = ICT perception.

Research question #1: How do individual factors influence the use of technology?

H1. *Teachers' level of technostress has a negative impact on their use of ICT.*

H2. *Teachers' digital pedagogical competence has a positive impact on their ICT use.*

H3. *Teachers' digital pedagogical competence has a negative impact on their technostress level.*

Research question #2: How do environmental factors influence individual factors and technology use?

H4. *University support has a negative impact on teachers' technostress levels.*

H5. *University support has a positive impact on teachers' digital pedagogical competence.*

H6. *University support has a positive impact on teachers' ICT use.*

H7. *Collegial support has a negative impact on teachers' technostress levels.*

H8. *Collegial support has a positive impact on teachers' digital pedagogical competence.*

H9. *Collegial support has a positive impact on teachers' ICT use.*

Research question #3: How do beliefs influence individual factors and the use of technology?

H10. *Teachers' ICT self-efficacy has a positive impact on their digital pedagogical competence.*

H11. *Teachers' ICT self-efficacy has a negative impact on their level of technostress.*

H12. *Teachers' ICT self-efficacy has a positive impact on their ICT use.*

H13. *Teachers' ICT perception has a positive impact on their digital pedagogical competence.*

H14. *Teachers' ICT perception has an indirect negative impact on their use of ICT.*

The definitions of the constructs we used are summarized in Table 1.

Table 1. Constructs and their definitions used in the research model.

Construct	Definition	Source
IU—ICT use	The frequency and extent of (self-reported) use of ICT for teaching and research purposes by higher education teachers.	[26]
TS—Technostress	„Any negative impact on attitudes, thoughts, behaviours, or body physiology that is caused either directly or indirectly by technology."	[18]
DC—Digital pedagogical competence	The combination of teachers' professional, pedagogical, and technological knowledge and skills.	[25]
US—University support	The human infrastructure and technological infrastructure component of teaching-focused ICT support originating from the institutional environment.	[32]
CS—Collegial support	The social component of teaching-focused ICT support originating from the institutional environment.	[32]
ISE—ICT self-efficacy	Teachers' belief and confidence in their ability to use technology effectively to achieve their educational goals.	[46]
IP—ICT perception	Teachers' perceptions of ICT show the extent to which teachers believe ICT is valuable in education.	[46]

4. Materials and Methods

4.1. Presentation of the Sample

The survey was conducted among university lecturers (n = 116) at the Károli Gáspár University of the Reformed Church in Hungary, with 6 different training locations and 5 faculties, using an online questionnaire (MS Forms). The sampled faculty members teach in 5 different disciplines (2 in biological sciences, 47 in philosophy and history, education, psychology, religious studies, 26 in economics and law, sociology, political science, 3 in mathematics, 35 in language and literature, 3 in medicine). The response rate was nearly 30%.

Our sample included 70 female and 46 male teachers. The youngest respondent was 27 and the oldest was 75. The average age was 49.68 years (SD = 10.00 years), while the average length of university teaching experience was 17.10 years (SD = 10.71).

4.2. Research Tool

The questionnaire used in this research consisted of two parts. The first part collected demographic information in the following areas: gender, age, number of years in education, and field of study. The second part consisted of seven subscales that measured each of the constructs, i.e., ICT use, technostress, digital pedagogical competence, university support, staff support, ICT self-efficacy, and ICT perceptions.

The survey was conducted using subscales already developed and validated by others and, thus, reliable, as well as reduced versions of these subscales, which were taken from research conducted in educational settings. After translating each item into Hungarian, an educational technologist was asked to validate the translation. Afterwards, we conducted a pilot survey with six other experts in the field of educational technology, and based on the problems and comments that arose, we improved the questionnaire from a linguistic and technological point of view.

ICT use and digital pedagogical competence were measured using a 6-point Likert scale, in line with the specificities of the DigCompEdu framework. The other questionnaire items were measured using the 5-point Likert scale (1: strongly disagree; 5: strongly agree).

To measure ICT use and Digital Pedagogical Competence, we used the higher education-specific version of the DigCompEdu survey [27], which consisted of 6-6 items. Items on ICT use were phrased with the verbs “use”, “support”, “create”, etc., which measured the extent to which a person does something, and, thus, differs from digital pedagogical competence, where respondents indicated how they rate their knowledge (“can do”) in different areas.

In this way, the use of technology measured actual use, while digital pedagogical competence measured knowledge, skills, and abilities.

The number of items and literature sources used to measure each variable are summarized in Table 2, while the individual items are shown in Table A3.

Table 2. Number of questionnaire items and literature sources.

Construct	Number of Items	Used Source
IU—ICT use	6	The higher education-specific version of the DigCompEdu survey [27], to measure knowledge, skills, and abilities.
TS—Technostress	3	“Technostress” subscale [60] (As the original scale referred to K-12 teachers’ technostress related to the use of mobile technology, “mobile technology” has been replaced with “ICT”, accordingly.)
DC—Digital pedagogical competence	6	The higher education-specific version of the DigCompEdu survey [27], to measure knowledge, skills, and abilities.
US—University support	3	“University ICT support” subscale [46].
CS—Collegial support	3	“Collegial support” subscale [5].
ISE—ICT self-efficacy	4	“Computer self-efficacy” subscale [5].
IP—ICT perception	4	“ICT perceptions” subscale [47] adapted from: [56].

4.3. Data Collection Process and Methods of Data Analysis

The electronic questionnaire on technostress among academics was open from the beginning of February to the end of May 2022 (data collection took place 1 year after the 2020 COVID-19 pandemic). The call was sent out three times directly to lecturers using their official work email address.

The collected data were analyzed using the SEM (structural equation modeling) model based on the PLS (partial least squares) procedure. The use of PLS–SEM path analysis was supported by the small sample size (n = 116) [61] and the fact that normality was violated [62].

The analysis was performed using SmartPLS 4.0.7.8 software [63]. The tool allows for the simultaneous construction of a confirmatory factor analysis (i.e., the measurement model) to test the reliability and validity of the constructs, and a path analysis (i.e., the structural model) to explore the relationships between the constructs and test the research hypotheses. In the modeling, path coefficients were estimated using a path weighting scheme. The fit of the structural model (testing the path coefficients) was checked by bootstrap sampling (5000 generated subsamples) using t-tests.

5. Results

5.1. Reliability Studies

To examine the reliability of the constructs, we used a Cronbach's alpha measure of internal consistency with a cut-off of 0.7 [64] and a composition reliability (CR) index, also with a cut-off of 0.7 [64,65]. As shown in Table A1, both conditions were met for all constructs, and the scales can, therefore, be considered clearly reliable.

Convergence validity was assessed using standardized factor weights and average variance extracted (AVE). The former was based on a criterion value of 0.6 as recommended by Schumacker and Lomax [66], and the latter on a criterion value of 0.5 as suggested by Fornell and Larcker [65]. The IP3 item was excluded from the analysis due to its factor weight of 0.588 in the IP variable. As can be seen in Table A1, all of the other standardized factor weights were above 0.6 and the mean extracted variances of the constructs were all above 0.5, thus, the convergence validity conditions were met.

The validity of discriminant validity [65] was verified using the correlation matrix and the average variance extracted (AVE) values (Table A2). Since the square root of the AVE values on the diagonal was greater than the off-diagonal correlation coefficients for all variables, discriminant validity between variables was satisfactory [65].

5.2. Testing the Hypothesized Model and Hypotheses

The model resulting from the PLS path analysis, with path coefficients and multiple coefficients of determination (R^2), is shown in Figure 2.

The results related to each hypothesis are listed below:

Technostress ($\beta = -0.096$, $p = 0.369$) has no significant effect on ICT use, so hypothesis H1 is rejected.

Digital pedagogical competence (DC) has a direct significant positive effect on ICT use (IU) ($\beta = 0.532$, $p < 0.001$), so hypothesis H2 is supported.

Digital pedagogical competence (DC) has a direct significant negative effect on technostress (TS) ($\beta = -0.250$, $p = 0.006$), so hypothesis H3 is supported.

Since the effect of university support (US) on technostress (TS) is neither direct ($\beta = -0.106$, $p = 0.115$) nor indirect ($\beta = 0.054$, $p = 0.536$), we reject hypothesis H4.

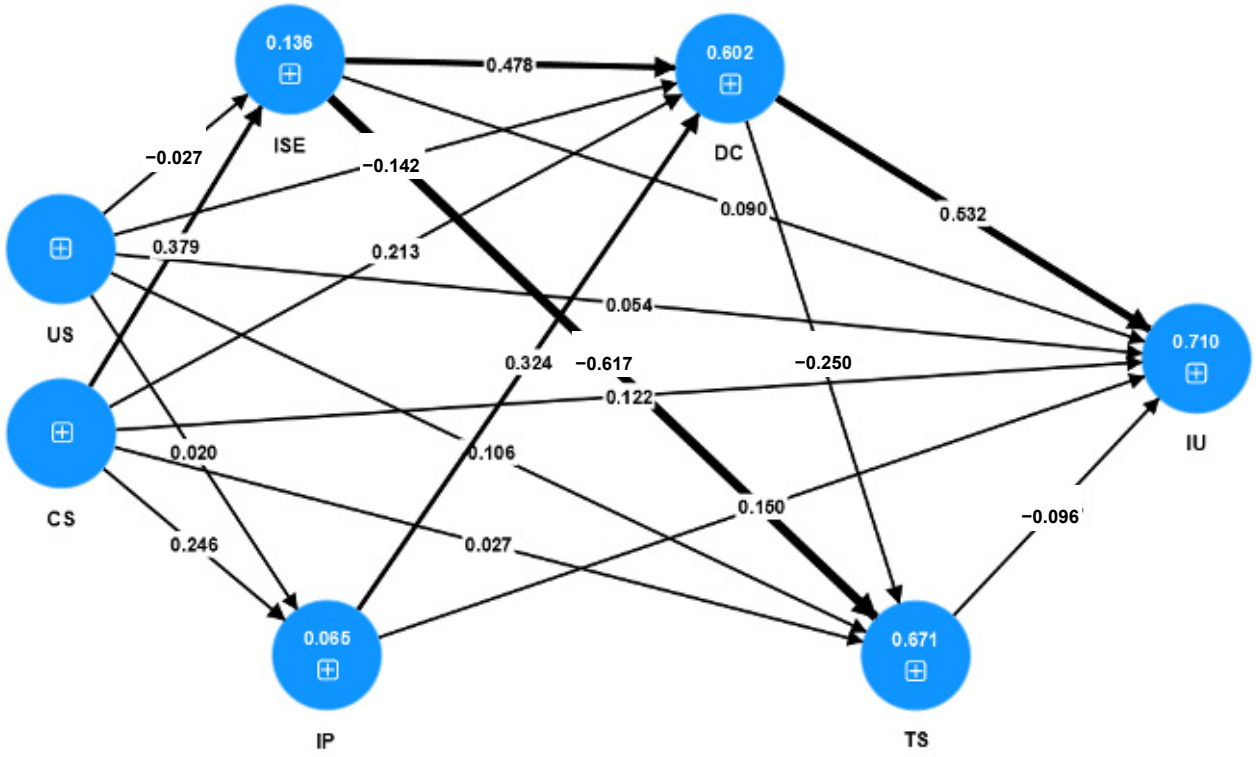


Figure 2. PLS results. Explained variance (R^2) is shown in the circles representing latent variables, path coefficients are shown as arrows. IU = ICT use; TS = technostress; DC = digital pedagogical competence; US = university support; CS = collegial support; ISE = ICT self-efficacy; IP = ICT perception.

As the direct effect of university support (US) on digital pedagogical competence (DC) is not significant (β = −0.142, p = 0.053), and the indirect effect is not significant either (β = −0.006, p = 0.932), we reject hypothesis H5.

Since there is no significant direct (β = 0.054, p = 0.439) or indirect (β = −0.073, p = 0.357) effect of university support (US) on ICT use (IU), we reject hypothesis H6.

Although the direct effect of collegial support (CS) on technostress (TS) is not significant (β = 0.027, p = 0.676), its indirect effect is (β = −0.353, p < 0.001), therefore, hypothesis H7 is partially supported.

Both the direct (β = 0.213, p < 0.001) and the indirect effect (β = 0.261, p < 0.001) of collegial support (CS) on digital pedagogical competence (DC) are significant, and, thus, hypothesis H8 is supported.

Although the direct effect of collegial support (CS) on ICT use (IU) is not significant (β = 0.122, p = 0.062), the indirect effect is (β = 0.354, p < 0.001), therefore, hypothesis H9 is partially supported.

ICT self-efficacy (ISE) has a significant direct positive effect on digital pedagogical competence (DC) (β = 0.478, p < 0.001), so hypothesis H10 is supported.

ICT self-efficacy (ISE) has a significant direct negative effect (β = −0.617, p < 0.001) on technostress (TS) and its indirect negative effect is also significant (β = −0.120, p = 0.004), so hypothesis H11 is supported.

Although the effect of ICT self-efficacy (ISE) on ICT use (IU) is not significant (β = 0.090, p = 0.434), its indirect effect is (β = 0.325, p < 0.001), therefore, hypothesis H12 is partially supported.

ICT perceptions (IP) have a positive direct effect on digital pedagogical competence (DC) (β = 0.324, p < 0.001), so hypothesis H13 is supported.

The positive direct effect of ICT perceptions (IP) on ICT use (IU) is significant (β = 0.150, p = 0.035), so hypothesis H14 is rejected.

The test model explains 71% of university teachers' ICT use (R2 = 0.710, adjusted R2 = 0.694).

6. Discussion

This study aimed to explore the factors influencing university teachers' use of ICT for teaching purposes and their structural relationships. We simultaneously investigated teachers' technostress and digital competence as individual factors, institutional support as an environmental factor, and teachers' beliefs, and the relationships and effects of these on ICT use.

Research question 1 aimed to examine how individual factors influence the use of technology.

In our study, technostress did not show a significant effect on ICT use, which was in contrast with previous claims that reducing teachers' technostress would increase their technology use [2,23,24]. This indicated that teachers' technology use does not necessarily depend on their level of technostress.

Among the individual factors, we examined the digital pedagogical competence of trainers, which is considered a critical factor in the literature.

Our analysis revealed that digital pedagogical competence (the strongest of the variables examined) had a significant positive effect on ICT use and a negative effect on technostress. This result replicated the findings of previous studies such as Kay [29], Joo et al. [2], and Dong [5].

On the one hand, this means that teachers' use of technology depends mostly on their current professional, pedagogical, and technological knowledge and skills. On the other hand, it also shows that teachers' level of competence in the use of technology for teaching purposes predicts their level of stress related to technology. Therefore, by improving teachers' professional, pedagogical, and technological knowledge and skills, their level of technostress can be reduced. Developing the necessary knowledge and skills requires long-term support [67]. It is, thus, essential to provide teachers with appropriate, long-term professional development support, which can take a variety of formats, from institutional face-to-face and group training through collegial, bottom-up self-study circles, to the provision of self-paced learning materials, all of which are already widely used in Hungarian higher education institutions [15].

Research question 2 considered how environmental factors influence individual factors and technology use. Of the environmental factors examined, university support was not shown to have a positive direct effect on influencing either technostress, digital pedagogical competence or technology use, nor was its indirect effect significant. This result was partially consistent with the findings of [5,46,47]

However, the role of collegial support was significant. On the one hand, it had both a significant direct positive impact on digital pedagogical competence and an indirect one, through its impact on ICT perception and ICT self-efficacy. The indirect effect of staff support on digital pedagogical competence was slightly stronger than its direct effect. This interesting correlation enhances the role of ICT perceptions and ICT self-efficacy in the development of teachers' digital competence.

On the other hand, the direct effect of collegial support on technostress and ICT use was not significant, indicating that peer support does not in itself reduce technostress and does not affect ICT use. Nonetheless, collegial support does have an indirect, significant and negative, effect on technostress and ICT use. By analysing these indirect effects, further interesting results were revealed:

Digital pedagogical competence played the largest mediating role in the indirect effect of collegial support on ICT use. Thus, staff support has a positive effect on ICT use mainly through influencing digital pedagogical competence. The largest mediator role of the indirect effect of staff support on technostress was ICT self-efficacy, i.e., collegial support can reduce technostress mainly by influencing ICT self-efficacy.

Overall, it can be concluded that improving the social infrastructure is a more effective way of developing teachers' digital pedagogical competences and reducing their technostress level than providing adequate human and technical infrastructure. A supportive atmosphere among teachers can be fostered by creating communities of practice so that teachers can develop their digital competence by providing emotional support to each other and sharing their knowledge. Joo [2] found similar results in their research.

In addition, by encouraging staff support, we can influence teachers' beliefs, thus, influencing their decisions on whether and how to use technology to support their learning and research processes, and also the level of technostress among teachers.

Finally, research question 3 asked how belief factors influence individual factors and technology use.

Both of the belief factors examined had a significant positive effect on digital pedagogical competence, with the effect of ICT self-efficacy being slightly stronger than the effect of ICT perception. Our results, similar to those of Taimalu and Luik [6], showed that teachers' existing beliefs and convictions about technology can influence the acquisition of knowledge. Taimalu and Luik [6] believed this to mean that if teachers feel technology is valuable, they will want to know more about it, and, thus, their digital pedagogical competence may be higher.

It is, therefore, important for institutions to demonstrate and emphasize, and for individuals to internalize, that ICT is valuable in education and can be used effectively to achieve educational goals.

Our results also showed that ICT self-efficacy negatively affects technostress. This means that teachers who believe that they can effectively use modern ICT tools to achieve their learning objectives have lower levels of psychological and physical stress related to technology use, i.e., anxiety and tension caused by potential problems and obstacles. In addition, even the indirect effect of ICT self-efficacy through digital pedagogical competence was significant.

Our further results showed that ICT perceptions have a positive direct effect on ICT use, a result also previously pointed out by Cheok et al. (2016, in connection with beliefs). That is, if instructors perceive technology as valuable, they are more likely to use it to facilitate learning.

We also examined the indirect effect of ICT perception on ICT use and found that technostress as a moderator was not significant; however, digital pedagogical competence as a moderator was significant, and the effect of ICT perception on ICT use through digital pedagogical competence was stronger than the direct effect itself. Thus, ICT perception has a positive effect on ICT use mainly through its influence on digital pedagogical competence.

Yet, a significant direct effect of ICT self-efficacy on ICT use has not been confirmed, similar to the findings of Taimalu and Luik [6] among teacher educators, and to Hickson's earlier research published in 2016 (although the latter examined self-efficacy in general in relation to ICT use). Therefore, teachers' belief in their ability to use technology effectively to achieve educational goals does not necessarily affect their actual use of technology. This could be explained, for example, by the fact that they believe that preparation is too time-consuming, which they believe does not "make up for" the benefits of ICT use.

For these reasons, it is important to introduce university teachers to the values of technology use (which can positively influence both their digital competence and their use of ICT). Taimalu and Luik [6] argued that, for example, we can increase university teachers' belief in the value of technology use by offering more meaningful training, tutorials and mentoring. In the same article, they also described that pedagogical beliefs are quite stable and long-term interventions or training are needed to change them [68]. Training, according to their research, provides knowledge but does not significantly change people's beliefs and attitudes towards technology [6].

7. Conclusions

7.1. Summary of Key Findings

In summary, we can conclude that digital pedagogical competence and collegial support, ICT self-efficacy, and ICT perception play a crucial role in technology use.

Considering the interplay of effects described in detail in the results section, this piece of research can contribute to the effective digitization of higher education in a practical way in several respects. Two areas of relevance are highlighted.

Our results showed that within institutional support, collegial support is of particular importance, i.e., encouraging communication, mutual cooperation and mutual support among colleagues is more effective in developing teachers' digital pedagogical competence than adequate infrastructure and individual technical support, and even has a positive impact on ICT use by influencing pedagogical competence.

Thus, the promotion of a supportive atmosphere among teachers can be seen as an important task for higher education institutions, but especially for centres of technological and methodological development. It is essential that, in addition to traditional course-based development, project-based, supportive forms of training should appear, embracing joint projects and bottom-up initiatives [69]. Furthermore, the development of communities of practice can be effectively supported and coordinated in connection with traditional training courses if we create the space for personal and professional contact, or if we try to build on the (professional) cooperation that has been established in courses offered at several levels/parts of training. Equally effective in this respect could be the targeted collection and joint training and support of different "interest groups" (e.g., (senior) management training, mentoring of new colleagues), or the collegial reflection, sharing and testing of good practices (even motivated by an institutional reward system [15]. In addition to the above, staff support also increases the ICT self-efficacy of teachers and, although not by itself, can effectively reduce technostress through its combined effect.

The other area worth mentioning based on the results is the area of beliefs about ICT tools. Beliefs strongly influence knowledge acquisition in the development of digital literacy.

ICT self-efficacy has a direct negative impact on technostress, while ICT perceptions have a positive impact on ICT use and contribute effectively to increasing digital pedagogical competences, thus, affecting ICT use at the same time.

However, it is interesting to note that ICT self-efficacy does not directly affect ICT use, so teachers' belief in their ability to use technology effectively to achieve learning objectives does not necessarily affect their actual technology use.

This is why it is important to demonstrate the value of technology use to university teachers: we can increase university teachers' belief in the value of technology use by offering more meaningful training, tutorials, and technology mentors. However, emphasising the usefulness of technology is only the first step: the most important thing is to strengthen teachers' belief in their own ability to use technology effectively and in a way that is relevant to their educational goals and objectives. This is, therefore, a much more complex task than the previous one, for which one of the possible means is to strengthen the appropriate use of self-reflection tools [26]. (For specific tools, see [69], chapter 1.3). Continuous feedback is also relevant in this context (e.g., originating from the course leader or participants at courses in methodological centre, or through awards and recognition).

As pedagogical beliefs (including those related to digital pedagogy) are quite stable, long-term interventions or training are needed to change them. Therefore, beyond training, universities need to create practical opportunities for teachers to experience the power of technology and how its use can help them in their daily work or in the organization of teaching-learning and other pedagogical processes.

There is a need for both expert mentors and more experienced colleagues to provide concrete examples of technology use, share best practices, and to provide active and continuous support and exchange of ideas on how to improve.

However, the hypothesis that either university support (human and technical infrastructure) or technostress play significant role in technology use was not confirmed in this

research, but we still consider both to be important factors that deserve attention in the long run.

7.2. Limitations and Further Directions of Research

As a limitation, we should mention first of all the small number of items, which suggests that similar studies should definitely be carried out on a larger sample, and more extensive data collection than at present would be needed to explore the correlations in depth.

Secondly, it is worth mentioning that alternative structural models may be possible, as path analysis cannot determine the direction of causality. This method only identifies the existence of correlation and the strength of the causal hypothesis but does not indicate the direction of causality.

Similarly, a self-reporting questionnaire can also be seen as a limitation, as respondents may under- or overestimate themselves, and, therefore, not necessarily measure the true state of the educator in relation to each area. It would be worthwhile to include a measure of actual usage and measure it against other factors in a research model. For a more complex investigation, additional methods are recommended: for example, observation can be used to obtain information on the integration of technology in education, and tests that can be used to measure digital competence can be used to collect real data on the knowledge of lecturers.

A fourth limitation is that data collection took place only one year after the digital remote education and, thus, after a particularly stressful period in terms of technology use, so it would be worthwhile to follow up the results by re-sampling the data or by further repeated measurements.

Finally, a limitation of the study is the convenience sampling at one university. Although the response rate was relatively high (around 30%), it would be necessary to include further national and even international universities, sampling in different contexts to generalize and refine the results and to better understand the factors of technostress.

Author Contributions: Conceptualization, J.T.N. and I.D.-H.; methodology, J.T.N.; software, J.T.N.; validation, J.T.N.; formal analysis, J.T.N.; investigation, J.T.N.; resources, J.T.N. and I.D.-H.; data curation, J.T.N. and I.D.-H.; writing—original draft preparation, J.T.N.; writing—review and editing, J.T.N. and I.D.-H.; visualization, J.T.N.; supervision, J.T.N. and I.D.-H.; project administration, J.T.N. and I.D.-H. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Ethical review and approval were waived for this study due to the following reason: All teachers from the University were invited to participate in the study by email communication, including a full description of the study in the form of a participant information sheet. Access permission was requested from the University Rector after approval by the Data Protection Commissioner. All teachers were informed that their participation in the online survey would be anonymous and voluntary and that they could quit the online survey whenever they wanted without suffering any disadvantage. All participants were informed that the results of the present study would be published after completion of the project. Informed consent was obtained from all individual participants included in the study.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The data presented in this study are available on request from the corresponding author. The data are not publicly available due their containing information that could compromise the privacy of research participants.

Conflicts of Interest: The authors declare no conflict of interest.

Appendix A

Table A1. The characteristics of constructs and indicators.

Construction	α	Items	Standardized Factor Loadings	CR	AVE
CS	0.866	CS1	0.824	0.916	0.785
		CS2	0.911		
		CS3	0.920		
DC	0.930	DC1	0.831	0.945	0.743
		DC2	0.858		
		DC3	0.904		
		DC4	0.819		
		DC5	0.903		
		DC6	0.854		
IP	0.795	IP1	0.892	0.878	0.707
		IP2	0.863		
		IP4	0.765		
ISE	0.899	ISE1	0.860	0.929	0.767
		ISE2	0.879		
		ISE3	0.888		
		ISE4	0.876		
IU	0.900	IU1	0.835	0.924	0.668
		IU2	0.763		
		IU3	0.844		
		IU4	0.809		
		IU5	0.846		
		IU6	0.804		
TS	0.782	TS1	0.865	0.872	0.694
		TS2	0.846		
		TS3	0.786		
US	0.739	US1	0.846	0.490	0.653
		US2	0.828		
		US3	0.750		

Notes: α = Cronbach-alfa; CR = composite reliability; AVE = average variance extracted; IU = ICT use; TS = technostress; DC = digital pedagogical competence; US = university support; CS = collegial support; ISE = ICT self-efficacy; IP = ICT perception.

Table A2. Construct correlations and the square root of AVE values.

Construction	CS	DC	IP	ISE	IU	TS	US
CS	0.886						
DC	0.408	0.862					
IP	0.255	0.594	0.842				
ISE	0.367	0.695	0.490	0.876			
IU	0.467	0.802	0.610	0.662	0.817		
TS	−0.349	−0.674	−0.639	−0.796	−0.676	0.833	
US	0.447	0.064	0.130	0.142	0.193	−0.198	0.809

Notes: In the correlation matrix the square root of average variance extracted (AVE) values are presented diagonally. IU = ICT use; TS = technostress; DC = digital pedagogical competence; US = university support; CS = collegial support; ISE = ICT self-efficacy; IP = ICT perception.

Table A3. Questionnaire items and literature sources.

Construct	Items	Used Source
TS	TS1. I feel tired from the workload through using ICT in teaching. TS2. I feel exhausted from using ICT for teaching. TS3. ICT activities make me feel stressed.	[60]
US	US1. Our university provides adequate ICT-related courses, which can meet my technology learning needs. US2. Our university provides sufficient information equipment, with which I can complete the technology-related tasks successfully. US3. I can obtain ICT-related support from my teachers or professional technicians easily.	[46]
CS	CS1. I received encouragement from my colleagues when I encountered difficulties in integrating ICT in teaching. CS2. Many colleagues shared useful resources and experience with me about integrating ICT in teaching. CS3. My colleagues and I made a concerted effort to integrate ICT in teaching.	[5]
ISE	ISE1. I can always manage to solve difficult problems in using ICT if I try hard enough. ISE2. It is easy for me to keep up with important new ICT. ISE3. I am confident that I have the technical skills to use ICT effectively. ISE4. When I am confronted with a problem when using ICT, I can usually find several solutions.	[5]
IP	IP1. ICT is very useful to me. IP2. The application of ICT makes teaching more effective. IP3. I always tend to apply new technology to specific content learning activities. IP4. ICT is a component of teachers' professional capability.	[47] adapted from: [56]

Notes: The answer options: 1: strongly disagree, . . . 5: strongly agree.

References

1. OECD. *Supporting the Digital Transformation of Higher Education in Hungary*; OECD Publishing, Higher Education: Paris, France, 2021. [CrossRef]
2. Joo, Y.J.; Lim, K.Y.; Kim, N.H. The effects of secondary teachers' technostress on the intention to use technology in South Korea. *Comput. Educ.* **2016**, *95*, 114–122. [CrossRef]
3. Dar, M.; Masood, F.; Ahmed, M.; Afzaal, M.; Ali, A.; Bibi, Z.; Kabir, I.; Zia, H.U. Information and communication technology (ICT) impact on education and achievement. In *Advances in Human Factors and Systems Interaction—Proceedings of the AHFE 2018 International Conference on Applied Human Factors and Ergonomics*; Nunes, I.L., Ed.; Springer: Cham, Switzerland, 2018; pp. 40–45. [CrossRef]
4. Tondeur, J.; Aesaert, K.; Prestridge, S.; Consuegra, E. A multilevel analysis of what matters in the training of pre-service teacher's ICT competencies. *Comput. Educ.* **2018**, *122*, 32–42. [CrossRef]

5. Dong, Y.; Xu, C.; Chai, C.S.; Ching, S.C.; Zhai, X. Exploring the Structural Relationship among Teachers' Technostress, Technological Pedagogical Content Knowledge (TPACK), Computer Self-efficacy and School Support. *Asia-Pac. Educ. Res.* **2020**, *29*, 147–157. [CrossRef]
6. Taimalu, M.; Luik, P. The impact of beliefs and knowledge on the integration of technology among teacher educators: A path analysis. *Teach. Teach. Educ.* **2019**, *79*, 101–110. [CrossRef]
7. Luik, P.; Taimalu, M.; Suviste, R. Perceptions of technological, pedagogical and content knowledge (TPACK) among pre-service teachers in Estonia. *Educ. Inf. Technol.* **2017**, *23*, 741–755. [CrossRef]
8. Amhag, L.; Hellström, L.; Stigmar, M. Teacher Educators' Use of Digital Tools and Needs for Digital Competence in Higher Education. *J. Digit. Learn. Teach. Educ.* **2019**, *35*, 203–220. [CrossRef]
9. Dringó-Horváth, I. IKT a tanárképzésben: A magyarországi képzőhelyek tanárképzési moduljában oktatók IKT-mutatóinak mérése. *Új Pedagógiai Szle.* **2018**, *9–10*, 13–41.
10. Hidalgo, F.J.P.; Parra, G.E.M.; Abril, C.H.A. Digital and media competences: Key competences for EFL teachers. *Teach. Engl. Technol.* **2020**, *20*, 43–59.
11. Madsen, S.S.; Thorvaldsen, S.; Archard, S. Teacher educators' perceptions of working with digital technologies. *Nord. J. Digit. Lit.* **2018**, *13*, 177–196. [CrossRef]
12. Bao, W. COVID-19 and online teaching in higher education: A case study of Peking University. *Hum. Behav. Emerg. Technol.* **2020**, *2*, 113–115. [CrossRef]
13. Guangul, F.M.; Suhail, A.H.; Khalit, M.I.; Khid-hir, B.A. Challenges of remote assessment in higher education in the context of COVID-19: A case study of Middle East College. *Educ. Assess. Eval. Account.* **2020**, *32*, 519–535. [CrossRef] [PubMed]
14. Fekete, I.; Divéki, R. The Role of Continuous Professional Development Workshops in the Technological-Pedagogical Skills Development of Teacher Trainers: A Case Study in the Hungarian University Context During COVID-19. In *Handbook of Research on Effective Online Language Teaching in a Disruptive Environment*; LeLoup, J., Swanson, P., Eds.; IGI Global: Hershey, PA, USA, 2022; pp. 201–220. [CrossRef]
15. Pintér, T.M.; Bodnár, É.; Dósa, K.; Dorner, H.; Lénárt, K.; Lengyelné, M.T.; Gorana, M.; Ollé, J.; Margaryta, R.; Vörös, Z.; et al. Oktatásinformatikai helyzetkép a magyarországi felsőoktatásban. *Új Pedagógiai Szle.* **2021**, *71*, 54–74.
16. Valverde-Berrocoso, J.; Acevedo-Borrega, J.; Cerezo-Pizarro, M. Educational technology and student performance: A systematic review. *Front. Educ.* **2022**, *7*, 916502. [CrossRef]
17. Brod, C. *Technostress: The Human Cost of the Computer Revolution*; Addison-Wesley Publishing Company: Reading, PA, USA, 1984.
18. Weil, M.M.; Rosen, L.D. *Technostress: Coping with Technology@ Work@ Home@ Play*; J. Wiley: New York, NY, USA, 1997.
19. Al-Fudail, M.; Mellar, H. Investigating teacher stress when using technology. *Comput. Educ.* **2008**, *51*, 1103–1110. [CrossRef]
20. Pamuk, S.; Peker, D. Turkish pre-service science and mathematics teachers' computer related self-efficacies, attitudes, and the relationship between these variables. *Comput. Educ.* **2009**, *53*, 454–461. [CrossRef]
21. Sabzian, F.; Gilakjani, A.P. Teachers' attitudes about computer technology training, professional development, integration, experience, anxiety, and literacy in English language teaching and learning. *Int. J. Appl. Sci. Technol.* **2013**, *3*, 67–75.
22. Voet, M.; De Wever, B. Towards a differentiated and domainspecific view of educational technology: An exploratory study of history teachers' technology use. *Br. J. Educ. Technol.* **2017**, *48*, 1402–1413. [CrossRef]
23. Maier, C.; Laumer, S.; Weinert, C.; Weitzel, T. The effects of technostress and switching stress on discontinued use of social networking services: A study of Facebook use. *Inf. Syst. J.* **2015**, *25*, 275–308. [CrossRef]
24. Suh, A.; Lee, J. Understanding teleworkers' technostress and its influence on job satisfaction. *Internet Res.* **2017**, *27*, 140–159. [CrossRef]
25. Koehler, M.J.; Mishra, P. What is technological pedagogical content knowledge (TPACK)? *Contemp. Issues Technol. Teach. Educ.* **2009**, *9*, 60–70. [CrossRef]
26. Dringó-Horváth, I.; Hülber, L.; Pintér, T.M.; Papp-Danka, A. A tanárképzés oktatási kultúrájának több szempontú jellemzése. In *Új Kutatások a Neveléstudományokban*; Varga, A., Andl, H., Molnár-Kovács, Z., Eds.; MTA Pedagógiai Tudományos Bizottság, PTE BTK Neveléstudományi Intézet: Pécs, Hungary, 2020; pp. 129–142.
27. Redecker, C.; Punie, Y. *European Framework for the Digital Competence of Educators, DigCompEdu*; Publications Office of the European Union: Luxembourg, 2017. [CrossRef]
28. Caena, F.; Redecker, C. Aligning teacher competence frameworks to 21st century challenges: The case for the European Digital Competence Framework for Educators (Digcompedu). *Eur. J. Educ.* **2019**, *54*, 356–369. [CrossRef]
29. Kay, R.H.; Knaack, L. An examination of the impact of learning objects in secondary school. *J. Comput. Assist. Learn.* **2008**, *24*, 447–461. [CrossRef]
30. Drossel, K.; Eickelmann, B. Teachers' participation in professional development concerning the implementation of new technologies in class: A latent class analysis of teachers and the relationship with the use of computers, ICT self-efficacy and emphasis on teaching ICT skills. *Large-Scale Assess. Educ.* **2017**, *5*, 19. [CrossRef]
31. Eickelmann, B.; Vennemann, M. Teachers' attitudes and beliefs regarding ICT in teaching and learning in European countries. *Eur. Educ. Res. J.* **2017**, *16*, 733–761. [CrossRef]
32. Nelson, M.J.; Voithofer, R.; Cheng, S.L. Mediating factors that influence the technology integration practices of teacher educators. *Comput. Educ.* **2019**, *128*, 330–344. [CrossRef]

33. Zhao, Y.; Pugh, K.; Sheldon, S.; Byers, J. Conditions for Classroom Technology Innovations. *Teach. Coll. Rec.* **2002**, *104*, 482–515. [CrossRef]
34. Fuglseth, A.M.; Sørebø, Ø. The effects of technostress within the context of employee use of ICT. *Comput. Hum. Behav.* **2014**, *40*, 161–170. [CrossRef]
35. Ragu-Nathan, T.; Tarafdar, M.; Nathan, R.; Tu, Q. The Consequences of Technostress for End Users in Organizations: Conceptual Development and Empirical Validation. *Inf. Syst. Res.* **2008**, *19*, 417–433. [CrossRef]
36. Salanova, M.; Llorens, S.; Cifre, E. The dark side of technologies: Technostress among users of information and communication technologies. *Int. J. Psychol.* **2012**, *48*, 422–436. [CrossRef]
37. Shin, W. Teachers' use of technology and its influencing factors in Korean elementary schools. *Technol. Pedagog. Educ.* **2015**, *24*, 461–476. [CrossRef]
38. Brown-Joseph, T.D. *A Study of the Barriers K–12 Teachers Encounter When Integrating Technology into the Curriculum*; University of Phoenix: Phoenix, AZ, USA, 2010.
39. Weber, D.M.; Kauffman, R.J. What drives global ICT adoption? Analysis and research directions. *Electron. Commer. Res. Appl.* **2011**, *10*, 683–701. [CrossRef]
40. Chen, R. Investigating models for preservice teachers' use of technology to support student-centered learning. *Comput. Educ.* **2010**, *55*, 32–42. [CrossRef]
41. Moreira-Fontán, E.; García-Señorán, M.; Conde-Rodríguez, Á.; González, A. Teachers' ICT-related self efficacy, job resources, andpositive emotions: Their structural relations with autonomous motivation and work engagement. *Comput. Educ.* **2019**, *134*, 63–77. [CrossRef]
42. Anderson, S.E.; Groulx, J.G.; Maninger, R.M. Relationships among Preservice Teachers' Technology-Related Abilities, Beliefs, and Intentions to Use Technology in Their Future Classrooms. *J. Educ. Comput. Res.* **2011**, *45*, 321–338. [CrossRef]
43. Ertmer, P.; Ottenbreit-Leftwich, A.T. Teacher Technology Change: How Knowledge, Confidence, Beliefs and Culture Intersect. *J. Res. Technol. Educ.* **2010**, *42*, 255–284. [CrossRef]
44. Voogt, J.; McKenney, S. TPACK in teacher education: Are we preparing teachers to use technology for early literacy? *Technol. Pedagog. Educ.* **2017**, *26*, 69–83. [CrossRef]
45. Tondeur, J.; van Braak, J.; Ertmer, P.A.; Ottenbreit-Leftwich, A. Understanding the relationship between teachers' pedagogical beliefs and technology use in education: A systematic review of qualitative evidence. *Educ. Technol. Res. Dev.* **2017**, *65*, 555–575, Erratum in *Educ. Technol. Res. Dev.* **2017**, *65*, 577. [CrossRef]
46. Liu, L.; Cruz, A.M.; Rincon, A.M.R. Technology acceptance, adoption, and usability: Arriving at consistent terminologies and measurement approaches. *Everyday Technol. Healthc.* **2019**, *23*, 319–338.
47. Wang, Q.; Zhao, G. ICT self efficacy mediates most effects of university ICT support on preservice teachers' TPACK: Evidence from three normal universities in China. *Br. J. Educ. Technol.* **2021**, *52*, 2319–2339. [CrossRef]
48. Bandura, A. Self-efficacy: Toward a unifying theory of behavioral change. *Psychol. Rev.* **1977**, *84*, 191–215. [CrossRef]
49. Salomon, G. Television is "easy" and print is "tough": The differential investment of mental effort in learning as a function of perceptions and attributions. *J. Educ. Psychol.* **1984**, *76*, 647–658. [CrossRef]
50. Blonder, R.; Jonatan, M.; Bardov, Z.; Benny, N.; Rap, S.; Sakhnini, S. Can You Tube it? Providing chemistry teachers withtechnological tools and enhancing their self-efficacy beliefs. *Chem. Educ. Res. Pract.* **2013**, *14*, 269–285. [CrossRef]
51. Yesilyurt, E.; Ulas, A.H.; Akan, D. Teacher self-efficacy, academic self-efficacy, and computer self-efficacy as predictorsof attitude toward applying computer-supported education. *Comput. Hum. Behav.* **2016**, *64*, 591–601. [CrossRef]
52. Byrne, B.M. *Structural Equation Modeling Using AMOS. Basic Concepts, Applications, and Programming*, 2nd ed.; Routledge: New York, NY, USA, 2010.
53. Rohatgi, A.; Scherer, R.; Hatlevik, O.E. The role of ICT self-efficacy for students' ICT use and their achievement in a computer and information literacy test. *Comput. Educ.* **2016**, *102*, 103–116. [CrossRef]
54. Admiraal, W.; Buijs, M.; Claessens, W.; Honing, T.; Karkdijk, J. Linking theory and practice: Teacher research in history and geography classrooms. *Educ. Action Res.* **2016**, *25*, 316–331. [CrossRef]
55. Semiz, K.; Ince, M. Pre-service physical education teachers' technological pedagogical content knowledge, technology integration self-efficacy and instructional technology outcome expectations. *Australas. J. Educ. Technol.* **2012**, *28*, 1248–1265. [CrossRef]
56. Baş, G.; Kubiatko, M.; Sünbül, A.M. Teachers' perceptions towards ICTs in teaching-learning process: Scale validity and reliability study. *Comput. Hum. Behav.* **2016**, *61*, 176–185. [CrossRef]
57. Mama, M.; Hennessy, S. Developing a typology of teacher beliefs and practices concerning classroom use of ICT. *Comput. Educ.* **2013**, *68*, 380–387. [CrossRef]
58. Knezek, G.; Christensen, R. Extending the will, skill, tool model of technology integration: Adding pedagogy as a new model construct. *J. Comput. High. Educ.* **2016**, *28*, 307–325. [CrossRef]
59. Avidov-Ungar, O.; Eshet-Alkalai, Y. Chais Teachers in a World of Change: Teachers' Knowledge and Attitudes towards the Implementation of Innovative Technologies in Schools. *Interdiscip. J. E-Learn. Learn. Objects* **2011**, *7*, 291–303.
60. Khlaif, Z.N.; Sanmugam, M.; Ayyoub, A. Impact of Technostress on Continuance Intentions to Use Mobile Technology. *Asia-Pac. Educ. Res.* **2023**, *32*, 151–162. [CrossRef]
61. Henseler, C.; Ringle, C.; Sinkovics, R. The Use of Partial Least Squares Path Modeling in International Marketing. *Adv. Int. Mark.* **2009**, *20*, 277–319.

62. Chin, W.W. The Partial Least Squares approach for structural equation modelling. In *Modern Methods for Business Research*; Marcoulides, A., Ed.; Lawrence Erlbaum Associates: Mahwa, NJ, USA, 1998; pp. 295–336.
63. Ringle, C.M.; Wende, S.; Becker, J.-M. SmartPLS 4, Boenningstedt: SmartPLS. 2022. Available online: https://www.smartpls.com (accessed on 22 August 2022).
64. Hair, J.F., Jr.; Black, W.C.; Babin, B.J.; Andersen, R.E.; Tatham, R.L. *Multivariate Data Analysis*, 7th ed.; Pearson Education: Upper Saddle River, NJ, USA, 2010.
65. Fornell, C.; Larcker, D.F. Structural Equation Models with Unobservable Variables and Measurement Error: Algebra and Statistics. *J. Mark. Res.* **1980**, *18*, 382–388. [CrossRef]
66. Schumacker, R.E.; Lomax, R.G. *A Beginner's Guide to Structural Equation Modeling*, 4th ed.; Routledge: New York, NY, USA, 2016.
67. Koh, J.H.L.; Chai, C.S.; Lim, W.Y. Teacher Professional Development for TPACK-21CL: Effects on Teacher ICT Integration and Student Outcomes. *J. Educ. Comput. Res.* **2017**, *55*, 172–196. [CrossRef]
68. Park, S.H.; Ertmer, P.A. Impact of Problem-Based Learning (PBL) on Teachers' Beliefs Regarding Technology Use. *J. Res. Technol. Educ.* **2007**, *40*, 247–267. [CrossRef]
69. Dringó-Horváth, I.; Dombi, J.; Hülber, L.; Menyhei, Z.; Pintér, T.M.; Papp-Danka, A. *Educational Technology in Higher Education—Methodological Considerations*; KRE ICT Research Center: Budapest, Hungary, 2021.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.